

19. Node Properties: type, tag aur contents

DOM Node Classes (hierarchy)

Har DOM node kisi built-in class ka hota hai, aur classes ek chain banati hain:

EventTarget -> Node -> Element -> HTMLElement -> (HTMLInputElement, HTMLBodyElement, HTMLAnchorElement, ...)

Aur Node -> Document, Node -> CharacterData -> (Text, Comment).

- EventTarget: events support ka base.
- Node: tree functionality (parentNode, childNodes, ...).
- Element: element navigation aur search (children, querySelector).
- HTMLElement: sab HTML elements ki base.

Kisi node ke properties = पूरी inheritance chain ka jod. Check karne ke tarike:

```
js
document.body.constructor.name;           // HTMLBodyElement
document.body instanceof HTMLElement;     // true
console.dir(elem); // properties dekhne ke liye (console.log tree dikhata hai)
```

nodeType (purana tarika, number deta hai)

- 1 = element, 3 = text, 9 = document.
- Read-only.

nodeName VS tagName

- tagName: sirf elements ke liye.
- nodeName: kisi bhi node ke liye (text ke liye #text, comment ke liye #comment, document ke liye #document).
- HTML me tag name hamesha UPPERCASE (BODY).

innerHTML

Element ke andar ka HTML string. Padh bhi sakte ho, badal bhi.

```
js
document.body.innerHTML = 'The new BODY!';
```

- Galat HTML browser theek kar deta hai.

- `<script>` insert karoge to **execute nahi hota**.
- `innerHTML += "..."` **append nahi, poora overwrite hai**: purana content hata kar naya likhta hai. Images reload hoti hain, input me likha text ud jaata hai, selection hat jaati hai.

`outerHTML`

Element ka poora HTML (khud element bhi). Dhyan: **write karne par element badalta nahi, DOM se hat kar naye HTML se replace ho jaata hai**. Purana variable purani value hi rakhta hai.

`nodeValue` / `data`

Text aur comment nodes ka content (`innerHTML` sirf elements ke liye hai). Aam taur par `data` use karte hain.

```
js
let text = document.body.firstChild;
text.data;
```

`textContent`

Element ka sirf **text**, saare tags hata ke.

- Padhna kam kaam aata hai.
- **Likhna bahut useful hai**: text ko safely daalta hai, tags ko literal text samajhta hai. User ke diye input ke liye ye safe hai (XSS se bachata hai).

```
js
elem2.textContent = "<b>hi</b>"; // literally "<b>hi</b>" dikhega
```

`hidden`

`elem.hidden = true` = `display:none` jaisa.

Aur properties (class ke hisaab se)

`value` (input/select/textarea), `href` (a), `id` (sab), `type` etc.

Tasks ke jawab (short)

- `document.body.lastChild.nodeType` script ke andar chalane par `1` (script khud aakhri node hai, baaki page abhi parse nahi hua).
- `body.innerHTML = "<!--" + body.tagName + "-->"` ke baad `body.firstChild.data` = `"BODY"`.

- `document` ki class `HTMLDocument` hai (chain: `HTMLDocument` -> `Document` -> `Node`).

Yaad rakho

`innerHTML` (HTML string, overwrite dhyan se), `outerHTML` (replace hota hai), `textContent` (safe text), `data` (text/comment), `nodeType`, `nodeName/tagName`, `hidden`.